

2014 JC2 H2 Biology Paper 3 (Suggested mark scheme)

Answer **all** questions.

- 1** A reporter gene is a gene whose activity is relatively easy to detect. Luciferase is an enzyme naturally found in fireflies. When its substrate luciferin is present, it catalyses a chemical reaction which emits light, as shown by the equation below. Therefore, luciferase gene can be used as a reporter gene in recombinant DNA technology.

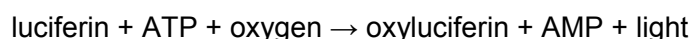


Fig. 1.1 shows a recombinant plasmid construct named “pMCS-Red Firefly Luc” used to study the regulation of *lac* operon in bacteria.

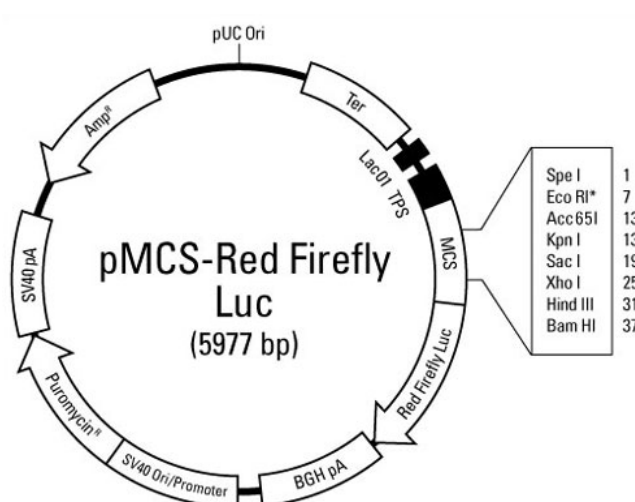


Fig. 1.1

Fig 1.2 shows the detailed structure of Red Firefly Luc insert.

promoter of <i>lac I</i> gene	<i>lac I</i> gene	CAP binding site	promoter	operator	luciferase gene
----------------------------------	-------------------	---------------------	----------	----------	--------------------

Fig. 1.2

- (a)** State the factors which you will need to consider when choosing the right [2]
restriction enzyme to construct the pMCS-Red Firefly Luc.
- The restriction site of the enzyme must be present in both the MCS of the plasmid and the sequence flanking the Red Firefly Luc insert, so that complementary sticky ends will be produced for ligation;;
 - The restriction site must only be found once in the whole plasmid / must be a unique cloning site;;

(b) Both genomic and cDNA libraries of firefly are available. State where you will take the luciferase gene from for this experiment. Justify your decision. [3]

- cDNA libraries;;
- Firefly is a eukaryote whose genes contain introns which need to be removed through splicing during post-transcriptional modifications of pre-mRNA;
- Bacteria are prokaryotes which do not perform post-transcriptional modifications;
- Hence, cDNA of luciferase gene which only contain exons;
- is needed for bacteria to produce functional luciferase

(c) Plasmids are introduced into bacteria via electroporation. [2]

Outline how you can select bacterial cells transformed with recombinant plasmid containing the Red Firefly Luc insert.

- Grow the bacteria in an agar medium containing ampicillin / puromycin, those surviving colonies are the ones transformed with plasmids;
- Add IPTG/ lactose (served as inducer to switch on the operon) and luciferin (substrate of luciferase) to the medium, those colonies which give out luminescence are cells transformed with recombinant plasmids containing the luciferase gene;

(d) A few experiments are conducted to study the regulation of *lac* operon via expression of luciferase gene in the bacterial colonies. [1]

Complete the table below by putting in '+' (low light intensity), '++' (high light intensity) and '-' (no light emitted) in the last column.

<i>experiment</i>	<i>glucose</i>	<i>lactose</i>	<i>ampicillin</i>	<i>luciferin</i>	<i>brightness of colonies</i>
1	✓	x	✓	✓	-
2	✓	✓	✓	✓	+
3	x	✓	✓	✓	++
4	x	✓	✓	x	-
5	x	x	✓	✓	-

(e) Explain the expected brightness of colonies for experiments 1 and 3. [4]

- In the absence of lactose, constitutively expressed lac I gene produces active repressor protein which binds to the operator, preventing RNA polymerase from transcribing the luciferase gene.
- No light is emitted when there is no luciferase catalysing the reaction in experiment 1.
- When lactose is present, allolactose acts as an inducer, which binds and changes the conformation of repressor protein. Thus the repressor protein can no longer bind the operator, allowing RNA polymerase to transcribe the luciferase gene;
- In the absence of glucose, there is high cAMP concentration in bacterial cell, leading to formation of active cAMP-CAP complex binding to CAP binding site, increasing the affinity of RNA polymerase for promoter and thus increasing transcription rate of the luciferase gene.
- High amount of luciferase causes more light to be emitted in experiment 3. (Combined marking point with 2nd bullet point)

(f) Outline the problems of cloning eukaryotic genes in prokaryotic cells. [2]

- Grow bacteria on medium containing ampicillin / puromycin to select those transformed with pMCS plasmid.
- To select colonies transformed with recombinant plasmid, add radioactively/fluorescently labeled single stranded nucleic acid probe complementary to the insulin cDNA sequence and perform autoradiography / view under UV light.

[Total: 14]

- 2 Cystic fibrosis is a genetic disease caused by mutations in the cystic fibrosis transmembrane regulator (CFTR) gene. An experiment utilising genetic engineering techniques was conducted to obtain genetically modified stem cells from monkeys suffering from cystic fibrosis.

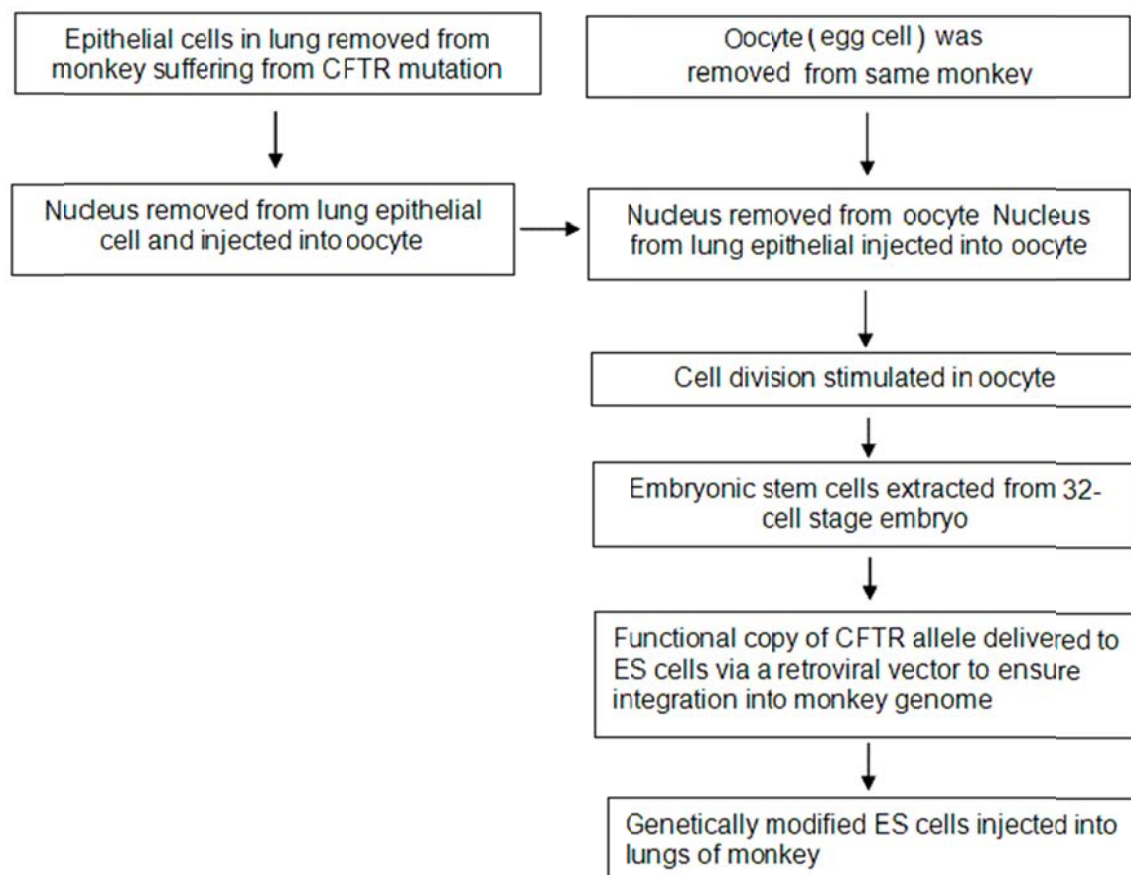


Fig. 2.1

(a) With reference to Fig. 2.1,

(i) describe the characteristics of the ES cells produced. [2]

- Self-renewal property: able to divide into 2 cells by mitosis, with one remain unspecialized while the other differentiate;
- Pluripotent : stem cells that can differentiate into almost any type of cell in the body;
- Can differentiate/specialize due to differential gene expression/ different sets of genes activated;

[max 2]

(ii) suggest why an oocyte was used and why the nucleus had to be removed from the oocyte. [2]

- oocyte is the only cell with potential to develop into a ball of cells/blastocyst in which all the cells are pluripotent;
 - haploid oocyte nucleus is removed so a diploid nucleus can be added/transferred;
 - Oocyte contains the organelles necessary for cell division and growth;
- [max 2]

(b) In another experiment, modified adenoviral vectors were used to deliver the functional copy of the CFTR allele into the nasal epithelium of monkeys with cystic fibrosis. Eight monkeys with cystic fibrosis were administered with different doses of the modified adenoviral vectors and empty adenoviral vectors. Each monkey was exposed to each type of vector once on the nasal epithelium.

Efficacy of gene transfer was determined six days after the administration of each respective vector. The results of the experiment are shown in Table 2.1.

Table 2.1

Monkey	gender / age	dose / plaque-forming unit (pfu)	presence of functional CFTR mRNA	
			treated with empty adenoviral vector	treated with modified adenoviral vector
1	Female / 25	2×10^7	No	No
2	Male / 44	2×10^7	No	No
3	Male / 23	2×10^8	No	No
4	Male / 21	2×10^8	No	No
5	Female / 34	2×10^9	No	Yes
6	Male / 40	2×10^9	No	No
7	Female / 19	2×10^{10}	No	Yes
8	Female / 19	2×10^{10}	No	Yes

plaque-forming unit (pfu): a measure on the quantity of virus particles

(i) Describe the main trends shown in Table 2.1. [2]

- No effects observed in individual 1 to 8 upon treated with empty adenoviral vectors, at a titre of 2×10^7 to 2×10^{10} pfu.
- Effect is observed at a high administration of modified adenoviral vectors, from 2×10^9 pfu.
- At a high titre of viral administration, effects was observed only in females but not male.

(ii) Suggest why the cystic fibrosis monkey was also treated with empty adenoviral vectors. [1]

- Serves as a control and basis of comparison for individuals treated with modified adenoviral vectors to show that empty adenoviral vectors do not express CFTR

Intervertebral discs sit between the vertebrae in the spine. They act as shock absorbers and allow movement of the spine. Over time, or following injury or disease, the discs wear out and can lead to back pain, either directly from the discs themselves or because of the pressure on the adjacent spinal cord.

Biomaterials are synthetic materials designed for use in the body. When used in conjunction with stem cells, they can act as a scaffold on which new tissue is formed. Biomaterials seeded with adult stem cells may hold the key to regenerating the damaged disc tissue and reducing pain. Two approaches are currently studied in laboratories. One approach is to produce whole, synthetic intervertebral discs seeded with stem cells. These synthetic discs look like normal discs and the cells grow on them. Alternatively, an artificial, thermosensitive hydrogel seeded with stem cells can be injected into the centre of a synthetic disc to form a fully functioning tissue. Over time, the artificial biomaterial disc and hydrogel biodegrade, leaving natural disc tissue produced by the cells themselves.

(c) Suggest why adult stem cells, rather than embryonic stem cells, are used [3]
when developing such stem cell-based therapies for the treatment of back pain.

1. The use of adult stem cells will allow doctors to use the patient's own stem cells for the therapy, rather than those from a donor.
 - This lessens the likelihood that the stem cells will be rejected by the patient's immune system.
2. There are fewer ethical concerns with the use of adult stem cells, particularly if they are the patient's own cells, than there are with embryonic stem cells.
3. Controlling the growth and differentiation of embryonic stem cells is more difficult than in adult stem cells, and may lead to incorrect tissue formation or uncontrolled cell growth.
 - This in turn can lead to the formation of cancerous tumours.

Max 3

[Total: 10]

3 One of the difficulties of culturing callus is surface sterilizing the piece of plant tissue (explants) to be used, especially when samples are being taken from the wild in tropical countries. In a study, plant tissue samples were taken at three different times of the year and placed on a medium containing no fungicide or antibiotic.

Two different sampling methods were used:

- discs cut from leaves wiped with bleach;
- stem tissues extracted with a sterile hypodermic syringe.

The results of culturing the explants are shown in **Table 3.1**.

Table 3.1

explant	time of year	number of explants	number of cultured explants with no fungal or bacterial contamination
leaf disc	April	153	12
	August	322	16
	January	332	30
stem tissue	April	194	116
	August	191	122
	January	211	156

(a) (i) Describe how explants may be grown into callus. [3]

- explants are transferred **aseptically**/ reference to **aspetic/ sterile condition** to prevent microorganisms growth/ use of laminar flow cabinet;
- to agar plate/ culture vessels / **culture medium**;
- containing nutrients such as a **sucrose as a carbon source**/ source of energy/ respiratory sunstrate,
- **organic nutrients**/ amino acids and vitamins **and inorganic ions**/ minerals/ named 2 e.g. nitrates, potassium, phosphates, etc.
- and **plant growth regulators**;
- of **intermediate ratio of auxin and cytokinin**;
- to stimulate growth to form callus tissue **by mitosis**;

(ii) With reference to **Table 3.1**, compare the effect of the different methods of taking tissue samples and the time of year on fungal and bacterial contamination of the cultured explants. [3]

- **stem tissue has less contamination** than leaf discs for all three times of the year;
- for example, in April, 60% of stem tissue has no fungal or bacterial contamination compared to only 8% of leaf discs or other appropriate comparison with manipulated data;;
- both treatments have **highest % non-contamination in January** of 74% for stem tissue and 9% for leaf discs respectively;;
- lowest for leaf discs in August of 5%, for stem tissue in April of 8%;;

(iii) Suggest a possible reason for the differences observed between the two types of explant. [2]

- leaf discs are **exposed to the external environment** and **bleach may not** be able to **remove all the contaminants** on them;;
- while using sterile hypodermic syringe to obtain the stem cells **directly** remove the sample hence the danger of **exposure to contaminants** in the external environment is **minimised**;;

Sulfonylurea which is a major component of a herbicide acts by binding to the enzyme acetolactate synthase (ALS), causing dysfunction of the enzyme and reducing the synthesis of certain amino acids that are necessary for protein synthesis. A mutation in the ALS gene found in weed was responsible for resistance to sulfonylurea.

An experiment was performed to study the effect of inserting the normal and mutated ALS gene from the weed into rice cells. The rice cells were grown in tissue culture and divided into four groups (**A**, **B**, **C** and **D**) according to their treatment as follows:

- A** : not genetically engineered
- B** : genetically engineered to express the normal ALS gene from weed
- C** : genetically engineered to express one copy of mutated ALS gene from weed
- D** : genetically engineered to express two copies of the mutated ALS gene from weed

Each group of cells was then grown into callus.

Fig. 3.1 shows the effect of different concentrations of sulfonylurea herbicide on the growth of each rice callus.

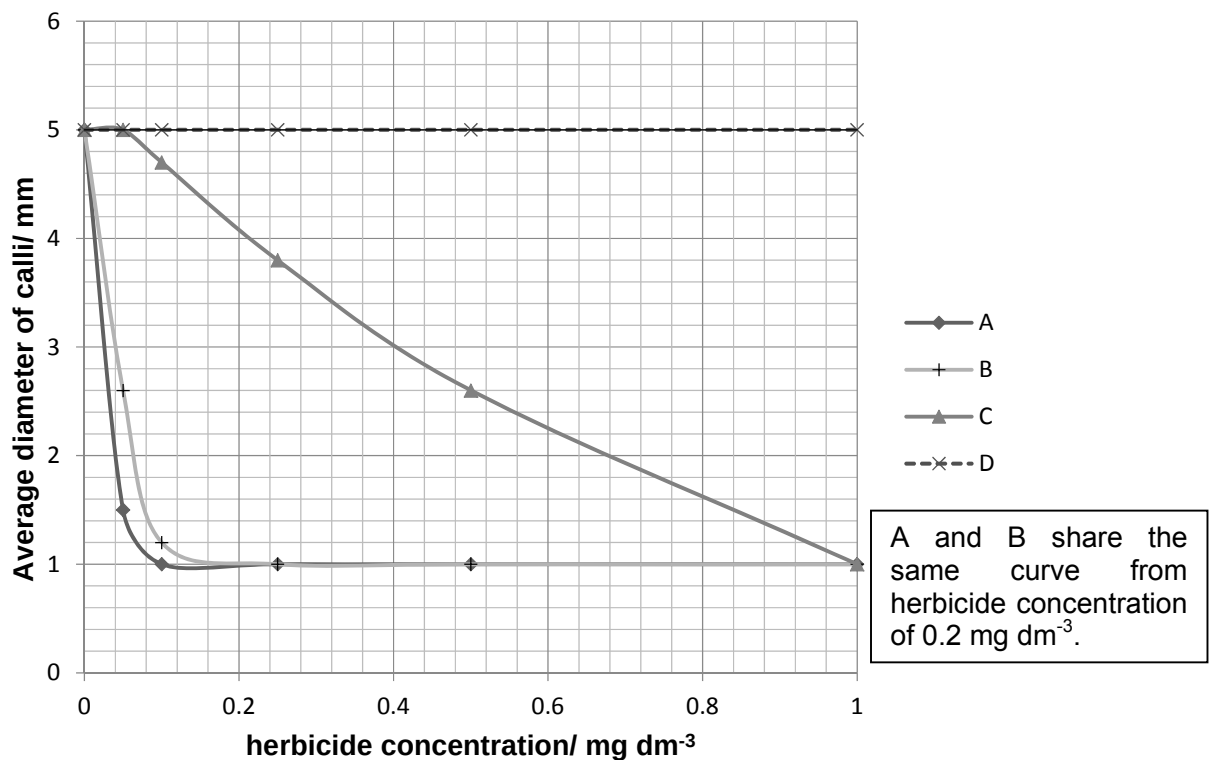


Fig. 3.1

(b) (i) Explain the purpose of calli **A** and **B** in this investigation. [2]

A: • Calli A grew minimally / grew the least at all concentrations of herbicide tested;

- to show that normal cells have minimal resistance/ do not have resistance to the herbicide;
- to serve as a base line for comparison of **C and D**;

[max 1]

B: • Calli B shows that the normal ALS gene from weed cannot cause herbicide resistance/ genetic engineering technique itself does not cause resistance (c.f. to **A**)

- so the results in **C and D** are due to the presence of mutation in gene from weed;;

(ii) State three advantages of using callus cells in producing genetically modified rice. [3]

1. Modified callus cells may be sub-cultured to obtain high multiplication rates
2. high reproductive capacity or high fecundity – get many plants within short time
3. genetic uniform – desirable traits can be pass down the progeny
4. idea of controlled microenvironment vs natural environment for plant growth - with examples
5. reasons for getting disease-free plants – benefits like high yield and robust, healthy plants
6. Undifferentiated modified callus cells may be induced to form plantlets;
7. Easy to obtain protoplasts to insert foreign DNA

[Any 3]

(iii) With reference to **Fig. 3.1**, assess the effect of mutated ALS gene on the expression of sulfonylurea resistance. Justify your answer. [3]

1. Presence of mutated ABL gene causes herbicide resistance (regardless of no. of copies) at low concentration of herbicide;
2. The concentration of herbicide that the cells can withstand is dependent / increases with the number of copies of the mutated ALS gene present/ reference to more copies of mutated ALS gene, greater resistance;
3. D has two copies of the mutated ALS gene is resistant even up to the high herbicide tested / 1.00mg/dm^{-3} ;
4. Average diameter of calli of D remain at 5 mm/ grow to maximum diameter of 5 mm at all herbicide concentrations tested
5. C has only one copy of the mutated ALS gene is only completely resistant to the herbicide up to concentrations of 0.05 mg dm^{-3}
6. After which the higher the herbicide conc. the greater the inhibition on callus growth of C as the size decreases from 5 mm to 3 mm at 0.4 mg dm^{-3} ,
(Points 5 and 6 mark for partially resistant to herbicide for one copy of mutated gene and QV)
7. Calli for C & D are larger than that for controls (A & B) at all concentrations of herbicide tested except at 1.00mg/dm^{-3} where C has a callus of the same average diameter as controls/ reference to comparison with controls to show that mutated ALS gene is responsible for sulfonylurea resistance;
8. Above 0.2 mg dm^{-3} , average diameter is 1 mm for A and B which is the size of the callus due to no resistance to herbicide;

[Total: 16]

4 Planning question

Cell division in flowering plants takes place in particular regions of the plant called meristems. Cells in meristems are not specialized for any particular function and divide repeatedly by mitosis. Some of the daughter cells remain meristematic; others cease dividing and become differentiated into appropriate cell types depending on their position.

Mitotic index is a measure for the proliferation status of a cell population. It is defined as the ratio between the number of cells in mitosis and the total number of cells. This can be done by observing cells undergoing cell division with a microscope.

You are required to plan an investigation on the difference in duration of each stage in the mitotic cycle, with respect to one another, in an actively dividing cell. This is to establish the longest stage in mitosis. In this planning, you will be provided with *Allium sativum*, commonly known as garlic.

Your planning must be based on the assumption that you have been provided with the following equipment and materials which you should use:

- a microscope
- *A. sativum* roots with intact tips
- toluidine blue which stains chromosomes blue
- 1M hydrochloric acid which fixes the specimen
- 70% ethanol
- Thermostatically water bath at 60 °C
- Scissor
- Watch glass (small glass dish)
- Beaker, 100 cm³,
- Scalpel
- Mounted needle
- Microscope slide
- Coverslip

Your plan should:

- include a clear statement of the hypothesis or prediction
- identify the key variables
- give full details and explanations of the procedures that you would adopt to ensure that the results are as precise and reliable as possible
- relevant, clearly labelled diagrams
- show how you would present and analyse your results
- include a brief risk assessment
- be written in clear scientific language, using the correct technical and scientific terms.

[Total: 12]

Theory (Max 2 marks)

1. Cell division in flowering plants takes place in particular regions of the plant called meristems.
2. Cells in meristems are not specialized for any particular function and divide repeatedly by mitosis.
3. Some of the daughter cells remain meristematic; others cease dividing and become differentiated into appropriate cell types depending on their position.
4. The root tip meristem is usually a denser white and more rounded than the cut end.
5. Chromosomes in root tip tissue are made visible with the stain.
6. Dividing cells (if present) will show up clearly with chromosomes in different forms according to the stage of mitosis.
7. Individual chromosomes (as tightly-coiled threads) are visible during anaphase.
8. The links between the cellulose walls of plant cells are broken down by the treatment with hydrochloric acid.
9. This ensures that the stain can penetrate the cells and allows the tissue to be squashed out one cell thick.
10. The mitotic index is calculated like this:

$$\text{Mitotic index} = \frac{\text{number of actively-dividing cells in field of view}}{\text{total number of cells in field of view}}$$

11. The duration of each stage of mitosis has been recorded and the data could be used to compare the observed frequencies of the different stages.
12. Hypothesis is that more cells at a stage, longer time taken to complete stage/ longest stage

Variables (max 2 marks)

Independent variables
stage of mitosis

dependent variable :
number of cells, which is used to calculate time taken to complete stage;

Award 1 mark for correct dependent and independent variables

Controlled variables :
use of same root tip
use of onions of similar stages
different root tips at similar development stage

Award 1 mark for at least 2 correct controlled variables.

Procedures (max 3 marks)**Preparation (max 2)**

13. Cut off 1-2 cm of the root tips. Put in a small volume of ethanoic acid on a watchglass (or other shallow dish) for 10 minutes.
Meanwhile, heat 10-25 cm³ of 1 M hydrochloric acid to 60 °C in a water bath.
14. Wash the root tips in cold water for 4-5 minutes and dry on filter paper.
15. Use a mounted needle to transfer the root tips to the hot hydrochloric acid and leave for 5 minutes.

16. Wash the root tips again in cold water for 4-5 minutes and dry on filter paper.
17. Use the mounted needle to remove two root tips onto a clean microscope slide.
18. Cut each about 2 mm from the growing root tip. Discard the rest, but keep the tips.
19. Add a small drop of stain and leave for 2 minutes.
20. Break up the tissue with a mounted needle.
21. Cover with a coverslip and squash using method A or method B below

Method A Place the slide and coverslip on a double layer of paper towel and fold the paper over the coverslip. Make certain that the slide is on a flat surface and squash down on the coverslip with a strong vertical pressure, using your thumb. Do not twist or roll the thumb from side to side

Method B Tap the coverslip about 20 times by dropping a wooden mounted needle or a pencil, blunt end down, from a height of about 5 cm onto the middle of the coverslip.

Investigation (max 1)

22. View the root tips under a microscope (x400 magnification) and look for the chromosomes within cells which are actively dividing.
23. Locate the meristematic zone, which has small, apparently 'square' cells with nuclei which are large relative to the whole cell area.
24. If cells are overlapping, squash the slide again. Avoid moving the coverslip from side to side.
25. Make sketches of (or take photographs of) cells that show any of the stages of mitosis.
26. Repeat step 22-24 in another field observation view (FOV) to increase reliability by obtaining 2 more readings.
27. Calculate the average of the three readings.
28. Calculate the mitotic index using the average of the three readings.

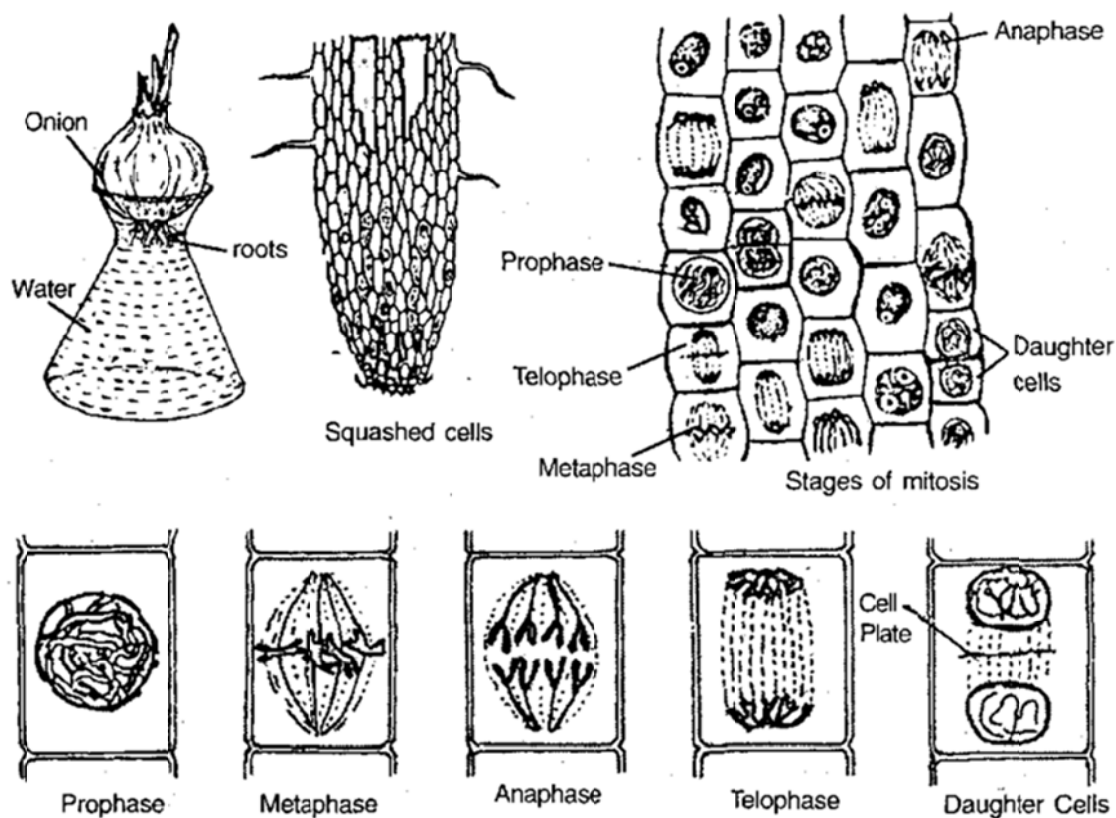
Data (Max 4 marks)

	Observed occurrence in representative microscope fields				
	number of cells				% of total cells (%)
	View 1	View 2	View 3	Average	
Prophase					
Metaphase					
Anaphase					
Telophase					

Award 1 mark for table of results with correct headers

Conclusion: The highest % indicates the longest stage of mitosis which is prophase.

Award 1 mark for conclusion.



Award 2 marks for large, labelled diagram of the different stages

Risk and precaution (max 1 mark)

1. Ethanoic alcohol is corrosive / skin irritant. Wear protective gear such as gloves when handling ethanol to avoid skin contact.
 2. Hydrochloric acid is corrosive / skin irritant. Wear protective gear such as eye protection when working with hydrochloric acid.
 3. AVP;;
-

[Total: 12]

Free-response question

- 5 (a) RFLP is a method used by molecular biologists to follow a particular sequence of DNA as it is passed on to other cells. RFLPs can be used in many different settings to accomplish different objectives. [8]

Explain how RFLP analysis facilitates the process of DNA fingerprinting and genomic mapping.

DNA fingerprinting:

1. DNA fingerprinting is a technology that **identifies particular individuals** using properties of their DNA;
2. Certain loci within human chromosomes contain **tandemly repeated sequences** (e.g. minisatellites) and the number of tandem repeats (at each minisatellite) varies substantially among individuals / is **highly polymorphic**;
3. Different number of tandem repeats leads to **generation of (different) restriction fragments** that are of different molecular weight and length giving rise to unique DNA fingerprint;
4. Each individual's combination is **unique**, having inherited one allele for each locus from each parent therefore DNA fingerprint of an individual is then analysed / compared against another individual to detect **matching band pattern**;
OR
5. Used for forensic analysis for crime investigation / paternity determination / trace evolutionary lineage;

Max 4

Genomic Mapping:

1. Genomic mapping is a **reconstruction of the entire set of chromosomes** for a given organism showing the relative position of every gene / genetic markers on a chromosome;
2. RFLPs greatly **increased the number of genetic markers available** for mapping the human genome (Reject: Gene markers / genome markers / marker genes;)
3. An RFLP serves as a **genetic marker** for a particular locus in the genome, (analogous to an allele of a particular gene) **OR** RFLPs are used as genetic markers to construct linkage map;
4. The frequency with which two RFLP markers or an RFLP marker and a certain allele for a gene is inherited together is a **measure of the closeness of the 2 gene loci** on a chromosome and the relative distance between 2 gene loci;;
5. Based on the recombination frequencies / frequency of crossing over between 2 RFLP markers / cross-over values of 2 RFLP markers, the linkage map is constructed **OR** The order of the genetic markers / the relative distances between genetic markers along a chromosome can be determined;

Any 4; max 4

- (b) The polymerase chain reaction (PCR) is arguably the most powerful [8]
laboratory technique ever invented.

Describe PCR and explain the advantages and limitations of this technique.

Stage 1:

1. DNA is separated into single strands by heating to **94°C** [accept 90 – 100°C] to **break of complementary hydrogen bonds**.

Stage 2:

2. Reaction mixture is cool to **55°C** [accept 50 – 60°C].
3. **Synthetic primers/ oligonucleotides** bind to their complementary sequences on either side of the singled-stranded target sequence at the 3' ends through the process of **annealing/hybridization**.

Stage 3:

4. Reaction mixture is heated to **72°C**; [accept 70 – 75°C] which is the **optimum temperature** (for *Taq* polymerase).
5. New strands (of DNA) are synthesised by **Taq polymerase** through the **addition of free deoxynucleoside triphosphates (DNTPs) to the free 3'OH** end of the primers by the formation of phosphodiester bond between the DNTPs.

Pt 1-5: 1 mark each (max 5 marks)

Advantages of PCR:

6. **Fast and efficient way** to amplify DNA with the exponential increase of the amount of DNA because the process of PCR can be automated;
7. PCR is **highly sensitive**. A target sequence can be amplified even when there is only a very small amount of the DNA available.

Max 1 mark for advantage

8. Give one useful example:

- Many copies of a gene of interest can be amplified for cloning purposes, as long as the primer sequences are known.
- Clinical diagnosis e.g. in prenatal diagnosis of human genetic disorders. DNA from single embryonic cells is amplified using PCR. Screening for certain genetic diseases e.g. cystic fibrosis even before the phenotype is expressed provides assurance to the couple. If child is diagnosed with certain diseases, couple can then be prepared and make informed choices.
- Early detection of infection with HIV: PCR can detect the presence of the HIV genome at very early stages of the diseases before symptoms appear.
- Forensic science: minute amount of DNA found e.g. at crime scene can be amplified to sufficiently large amounts to be analysed.
- Study of fossil: PCR is used to amplify fragments of ancient DNA from a 40,000 year old frozen woolly mammoth.

1 mark for example

Limitation of PCR:

9. Taq polymerase **lacks the 3' to 5' proofreading mechanism**. This makes it impossible for the enzyme to check the base it has just inserted and remove it if it is incorrect. Polymerase induced errors may accumulate during PCR.
10. PCR **requires at least the knowledge of the sequences that flank the target DNA** to synthesis the primers. Without proper primers, then PCR cannot take place.
11. **DNA fragments are limited to 3 kb**. Further increase in length will decrease the efficiency of amplification.

Max 1 mark for limitation

- (c) The Human Genome Project (HGP) was one of the great feats of exploration in history - an inward voyage of discovery using techniques of RFLP and PCR. [4]

Describe the goals of the human genome project.

1. To **determine the sequence of the coded information** contained in the DNA of the various genomes studied and from this to identify all of the estimated 30,000 human genes;
2. **Constructing a detailed physical map** of the entire human genome;
3. **Functional genomics**: studying genes' normal functions, the extent to which they vary from person to person and the role of this variation in causing or contributing to developmental, growth and health problems;
4. Storing all sequence information in **databases** that are accessible by all;
5. Studying and addressing the **ethical, legal and social implications** (ELSI) of genome research;
6. **Improving tools for analysis** such as sequencing technology development, developing technology for functional genomics, developing technology in bioinformatics;
7. **Completing the sequences of several other organisms** such as bacteria, yeast, *Drosophila melanogaster* and the mouse, to be used as models for research. Knowledge gained by the study of genomes of other organisms would assist in the analysis of the human genome;

Any **4****The End**